

TT-GAN: A Hydroponic *Talinum triangulare* Image Dataset and an Evaluation of VAE-GAN Hybrid Performance Under Limited Data

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Abstract:

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1. INTRODUCTION

Agricultural productivity and food security remain critical challenges in many developing regions where indigenous crops play a central role in nutrition and income. Despite their importance, many of these crops remain underrepresented in modern data driven agricultural research, especially in the context of machine learning and generative modelling [1]. This lack of datasets limits the application of artificial intelligence (AI) techniques for growth analysis, simulation and decision making in digital farming systems. Recent advances in deep learning generative models, particularly Generative Adversarial Networks (GANs) and Variational Auto encoders (VAEs), have demonstrated capabilities in learning complex data features and generating realistic samples across domains such as computer vision, medical imaging and simulation environments [2, 3]. However, the majority of existing applications rely on large datasets derived from controlled laboratories, industrial farms or non-indigenous crop varieties. As a result, the potential of generative modelling remains untouched for indigenous African crops grown under locally relevant conditions.

Talinum triangulare, commonly known as waterleaf, is a widely consumed leafy vegetable across West and Central Africa. It is an herbaceous plant widely grown in tropical regions as a leaf vegetable with high moisture content of almost 90.8 g per 100 gm of edible leaf [4]. It is an erect herb with swollen roots and succulent stems and 30-100 cm tall. The branches have two lateral basal buds. The leaves are spirally arranged nearly opposite and often crowded at the top of the stem. The plant is valued for its high moisture content, rich mineral composition and nutritional benefits including significant levels of vitamins, antioxidants and dietary fibre [6]. It is valued for its rapid growth cycle, nutritional content and adaptability to diverse environments [4]. Despite its agricultural and socioeconomic relevance, publicly available datasets capturing its growth dynamics, particularly under controlled and sensor-monitored conditions, are non-existent. This data gap presents a significant barrier to the development of AI-driven tools for simulation, monitoring and optimization of its cultivation.



Figure 1: *Talinum triangulare* [5]

Hydroponic farming systems offer a controlled growth environment that is well suited for data acquisition. By minimizing soil-related variability and enabling regulation of water availability, hydroponics provides a repeatable experimental setup for studying plant growth patterns [9]. When integrated with Internet of Things (IoT) sensors, such systems enable continuous monitoring of environmental parameters, improving data reliability and reproducibility. In particular, water level sensors play a critical role in hydroponic setups as water availability directly influences plant development.

Generative modelling techniques have evolved beyond single architectures by moving towards hybrid architectures that combine the strengths of multiple models. Variational Autoencoders offer structured latent representations and stable training but often suffer from blurry outputs while Generative Adversarial Networks are capable of producing clear images but may experience training instability [11]. Hybrid VAE-GAN architectures address these limitations by jointly learning meaningful latent spaces and adversarial realism thus making them well suited for simulation tasks where explanation and clarity is needed.

Building on these observations, this work introduces TT-GAN, a dataset and image synthesis framework centered on the hydroponic cultivation of *Talinum triangulare*. The proposed framework integrates an IoT-monitored hydroponic system for reproducible image acquisition with a VAE-GAN based generative model designed to simulate plant growth patterns. In contrast to prior studies that primarily emphasize generative model performance, this work puts forward dataset development as its core research contribution.

The contributions of this work are as follows:

- 1) To introduce a novel and publicly available hydroponic *Talinum triangulare* image dataset captured under controlled growth conditions.
- 2) To conduct an investigation of how limited dataset size affects the visual fidelity of VAE-GAN hybrid architectures.
- 3) To provide qualitative and quantitative analysis highlighting architectural and data-driven limitations in biologically complex plant image synthesis.

2. LITERATURE REVIEW

Talinum triangulare, commonly referred to as waterleaf, is a fast-growing leafy vegetable widely cultivated and consumed across West Africa. Due to its short growth cycle and adaptability, *Talinum triangulare* plays an important role in household food security and small-scale farming systems. Despite its agricultural relevance, existing literature on *Talinum triangulare* are largely limited to nutritional analysis, traditional cultivation methods and medicinal properties [7, 8]. There is a notable absence of publicly available datasets documenting its growth patterns. This lack of structured data limits the application of machine learning and computer vision for this crop.

Hydroponic farming is a soil-less cultivation method in which plants are grown using water solutions. By eliminating soil-related variability, hydroponics enables precise control of environmental parameters such as water availability, nutrient concentration and growth conditions [9]. These characteristics make hydroponic systems particularly suitable for experiments and reproducible data collection. In recent years, hydroponic systems have been increasingly adopted in controlled environment agriculture particularly for research purposes [12]. Studies have demonstrated that hydroponics can enhance growth consistency and facilitate continuous monitoring when integrated with sensors [13].

Variational Autoencoders are probabilistic generative models that learn latent representations of data through variational inference [3]. Unlike normal autoencoders, VAEs arrange their hidden features in an orderly way. This makes it easy to smoothly move between data points and create new data. VAEs have been applied across a range of domains including image synthesis, anomaly detection and data modelling [16, 17]. However, VAEs often have a problem as the images they generate can look too smooth or blurry, especially when the input images are complex [14]. This limitation has motivated the

development of hybrid architectures that combine VAEs with adversarial learning mechanisms to improve output realism [15].

Generative Adversarial Networks, introduced by Goodfellow et al. [2], consist of a generator and a discriminator trained in an adversarial framework. GANs have demonstrated success in generating high-quality images and handling complicated data patterns, leading to widespread adoption in generative AI applications. Despite their strengths, GANs are known to have problems during training such as training instability, producing limited types of outputs and being very sensitive to hyperparameters [18, 19]. Numerous variants have been proposed to address these challenges, including conditional GANs, Wasserstein GANs and architectural refinements [20-22]. GANs excel at making realistic images but their hidden features are not well organized. This makes it harder to understand them and to control the images they generate [23-25]. An image of a basic Generative Adversarial Network is illustrated in Figure 2 describing the interaction of Generator and Discriminator in adversarial learning where the Discriminator aims to predict real or generated images.

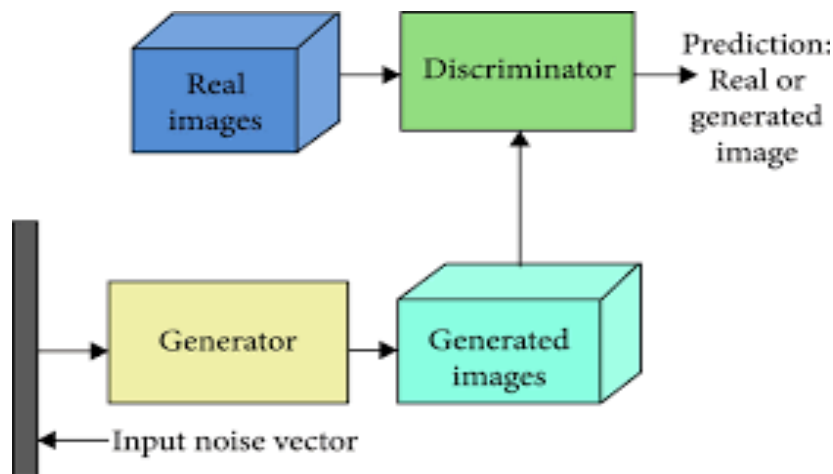


Figure 2: Basic Diagram of Generative Adversarial Network [26]

Hybrid architectures combining VAEs and GANs have been proposed to combine the strengths of both models. One of the earliest and most influential approaches was the VAE-GAN framework introduced by Larsen et al. [11], which integrated adversarial training into the VAE reconstruction process. This approach enabled the learning of structured latent representations while improving the perceptual quality of generated samples. Subsequent studies tried combining VAEs and GANs in different ways. These hybrids showed better training stability, produced more diverse outputs and created more realistic images than using VAEs or GANs alone [27-29]. These properties make VAE-GAN architectures particularly suitable for simulation tasks, where both consistency and realistic output are required. However, their application to agricultural image generation remains largely unexplored.

The integration of Internet of Things (IoT) technologies into agriculture has enabled real-time monitoring and improved resource management [30, 31]. IoT-based agricultural systems typically employ sensor networks to collect environmental data such as temperature, humidity and water level [32]. In hydroponic setups, sensors play a crucial role in ensuring stable growth conditions and gathering datasets. Water-level monitoring is essential for maintaining adequate water availability and preventing stress conditions that may affect plant development [33]. IoT-enabled monitoring improves data traceability which is crucial for research.

High-quality datasets are a foundational requirement for the development and evaluation of machine learning models. In agricultural AI, the majority of publicly available datasets focus on plant disease detection, yield estimation or large-scale commercial crops [34]. Indigenous crops and small-scale farming systems remain underrepresented particularly in developing regions. Several studies have highlighted data scarcity as a major barrier to AI adoption in agriculture across Africa and other developing regions [35, 36]. Liang et al. [37] highlights the importance of careful dataset design and curation in building trustworthy AI systems emphasizing strategies to ensure data quality and diversity while addressing biases and ethical considerations. The publication of a domain specific dataset has therefore been recognized as valuable research contributions in their own right.

The reviewed literature highlights significant advances in generative modelling, controlled environment agriculture and IoT-based monitoring systems. However, a clear gap exists at the intersection of these domains. Existing works rarely combine indigenous crop datasets, sensor-aligned hydroponic growth monitoring and hybrid generative models within a

unified framework. In particular, no publicly documented dataset or generative image framework currently exists for the hydroponic growth of *Talinum triangulare*. This work addresses these gaps by introducing a novel dataset and VAE-GAN-based image synthesis framework developed in IoT-monitored hydroponic cultivation, thereby extending generative modelling research into an under explored domain.

3. METHODOLOGY

This section explains the full experimental process employed in this study, including the hydroponic setup, the IoT monitoring system, how the dataset was collected and prepared and how the VAE-GAN model was designed and trained.

A. Overview of the Experimental Pipeline

The methodology followed a four-stage pipeline:

- 1) Controlled hydroponic cultivation of *Talinum triangulare* using stem cutting.
- 2) IoT-based monitoring of water level using a water-level sensing system.
- 3) Image dataset acquisition and curation building a structured hydroponic growth dataset.
- 4) Development and training of a VAE-GAN architecture for growth pattern simulation.

An overview of the complete system architecture is illustrated in Fig. 3.

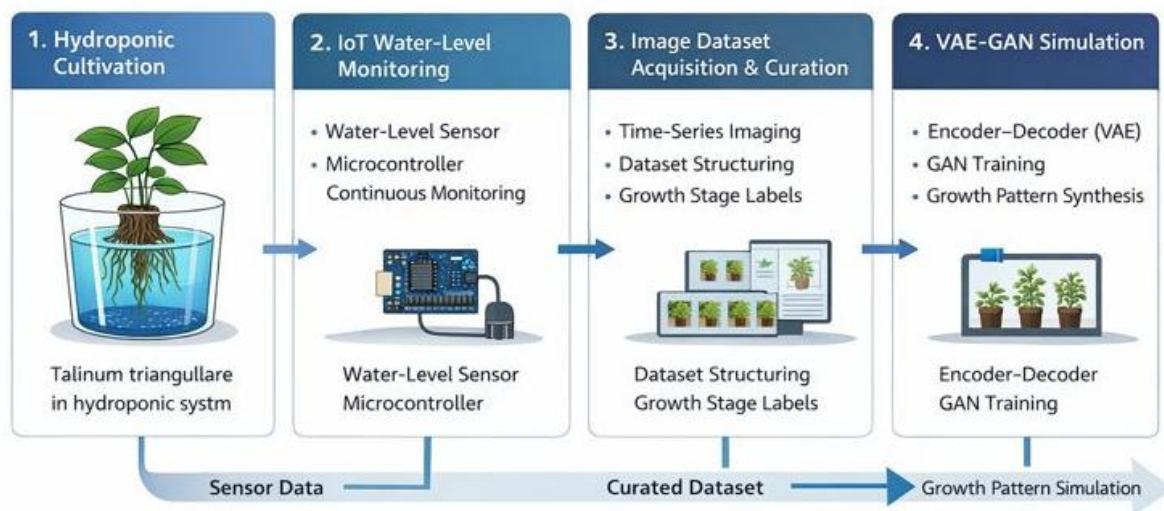


Figure 3: System Architecture

B. Hydroponic Growth Setup for *Talinum triangulare*

1) Plant Material and Propagation Method: *Talinum triangulare* was propagated using a stem cutting obtained from a mature and healthy parent plant. Stem cutting was selected due to its widespread use in traditional cultivation and suitability for rapid propagation. The stem cutting measured 14 cm in length with a distance of 10.3 cm from basal end of stem to the first node. Multiple nodes were retained to ensure successful rooting and early growth initiation. The cutting was allowed a healing delay of 60 minutes post cutting to minimize the risk of rot. Prior to placement in the hydroponic system, the cutting was visually inspected to ensure no damage. No chemical rooting agents were applied, allowing the growth process to reflect natural propagation behaviour. Figure 4 illustrates the freshly prepared stem cutting, highlighting the presence of seven (7) nodes.



Figure 4: Stem Cutting of *Talinum triangulare*

2) Hydroponic System Design: The system consisted of a transparent container measuring 12 cm in length partially filled to 10 cm with tap water, supporting the plant stem such that the basal end was submerged while the leaves remained above the water surface. This enabled direct observation of root development and allowed consistent water exposure. The use of a transparent container also allowed visual inspection and image capture without disturbing the plant. This is illustrated in Figure 5.



Figure 5: Hydroponics setup

C. IoT-Based Water-Level Monitoring System

1) Sensor Selection and Hardware Components: To monitor water availability within the hydroponic system, a water-level sensor was integrated into the setup. The sensor was interfaced with an Arduino Uno R3 microcontroller which served as the central data acquisition and processing unit. The sensor measured changes in water level and based on a threshold value of 600, activated a visual indicator where the green LED illuminated when the reading was at 600 and above, while the red LED was triggered for readings 599 and below. Simultaneously, the exact water level was displayed in real time on an LCD screen.

Primary hardware components included:

- a) Arduino Uno R3 microcontroller
- b) Water-level sensor
- c) Battery pack with four AA batteries
- d) Green and red LEDs
- e) $220\ \Omega$ and $10\ \text{k}\Omega$ resistors
- f) Male-to-Male and Male-to-Female Jumper wires
- g) 16x2 LCD I2C

The water-level sensor was positioned vertically at a slight angle within the hydroponic container maintaining a safe distance from the plant stem to avoid interference with growth. The sensor plates were fully submerged while the electronic components remained above the water level preventing water damage of the sensor. The wiring diagram of the hydroponic monitoring system, illustrated in Figure 6, depicts the interconnection of all system components. The water-level sensor is connected to the analog input pin of the Arduino Uno R3. The green and red LEDs are connected to digital output pins through $220\ \Omega$ current limiting resistors enabling the Arduino to control the visual indicators based on sensor readings. The LCD display is interfaced with the Arduino using I2C communication lines presenting real-time water-level readings as output. Power is supplied to the Arduino from a standard 5V source which powers the system.

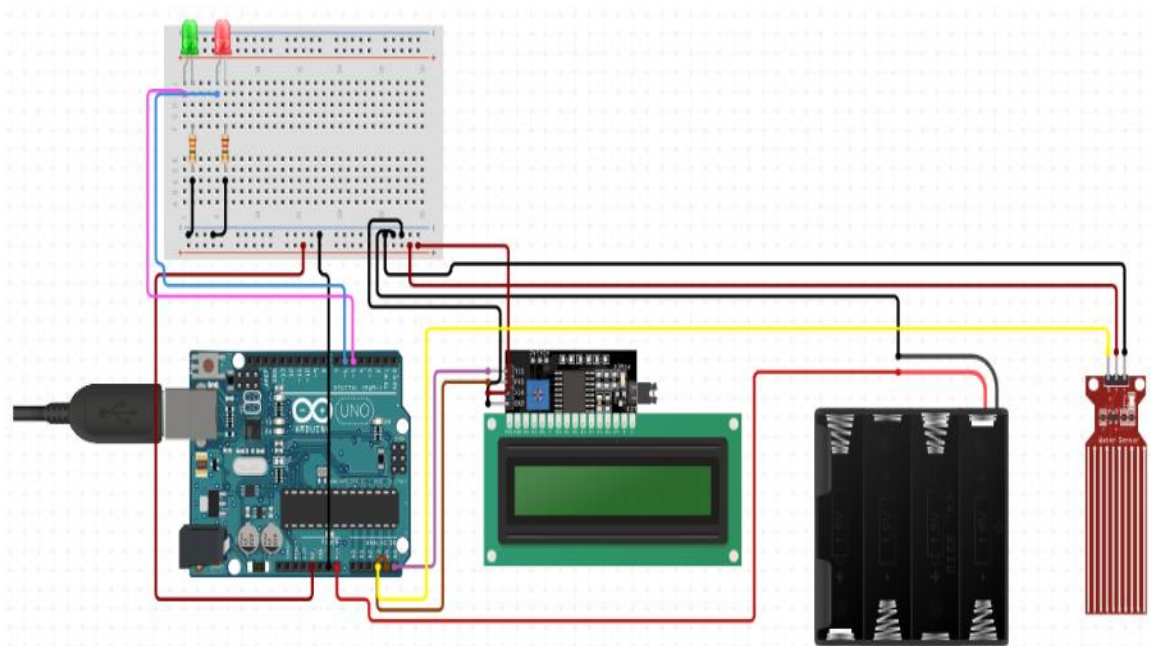


Figure 6: Wiring Diagram for the IOT Based Water Monitoring System

2) Data Acquisition and Logging: The microcontroller monitored the water level sensor with readings transmitted via a serial connection to a connected computer for real-time recording. Each measurement was logged with a timestamp enabling continuous tracking of water-level fluctuations throughout the plant growth period. The recorded data were regularly reviewed to verify sensor operation.

A sample serial monitoring output is shown in Figure 7.

```

10:58:46.032 -> LCD detected at 0x27
10:58:47.152 -> Water: 527, Status: GOOD
10:58:49.210 -> Water: 529, Status: GOOD
10:58:51.267 -> Water: 526, Status: GOOD
10:58:53.325 -> Water: 527, Status: GOOD
10:58:55.384 -> Water: 528, Status: GOOD
10:58:57.440 -> Water: 526, Status: GOOD
10:58:59.495 -> Water: 521, Status: GOOD
10:59:01.550 -> Water: 523, Status: GOOD
10:59:03.608 -> Water: 521, Status: GOOD
10:59:05.664 -> Water: 520, Status: GOOD
10:59:07.681 -> Water: 523, Status: GOOD
10:59:09.737 -> Water: 520, Status: GOOD
10:59:11.801 -> Water: 521, Status: GOOD
10:59:13.858 -> Water: 525, Status: GOOD
10:59:15.920 -> Water: 522, Status: GOOD
10:59:17.983 -> Water: 524, Status: GOOD
10:59:20.042 -> Water: 516, Status: GOOD
10:59:22.053 -> Water: 514, Status: GOOD
10:59:24.110 -> Water: 513, Status: GOOD
10:59:26.201 -> Water: 510, Status: GOOD
10:59:28.211 -> Water: 518, Status: GOOD
10:59:30.269 -> Water: 516, Status: GOOD
10:59:32.321 -> Water: 513, Status: GOOD
10:59:34.383 -> Water: 513, Status: GOOD
10:59:36.442 -> Water: 524, Status: GOOD
10:59:38.499 -> Water: 520, Status: GOOD
10:59:40.550 -> Water: 525, Status: GOOD

```

Figure 7: Serial Monitor Output from Arduino Software.

D. Image Dataset Acquisition and Curation

1) Image Capture Protocol: Images of *Talinum triangulare* were captured daily at regular intervals throughout the hydroponic growth period using the rear and front camera of a smartphone. Image acquisition was conducted under natural daylight conditions indoors between the hours of 0800 and 0900 with consistent camera positioning to minimize variability unrelated to plant growth. The camera distance was kept as uniform as possible with a fixed background across all capture sessions to ensure consistency. Image capture was performed over a period of 30 days with photographs taken from four viewpoints: front view, top view and two side views. This multi-angle strategy aided to capture visual information on plant morphology and development over time. In total, 120 images were collected and curated to form the final growth dataset.

2) Dataset Scope and Structure: Each image was carefully checked to make sure it was clear, useful and properly framed. Images with blur, blocked views or poor lighting were removed from the dataset. Representative dataset samples are shown in Figure 8.

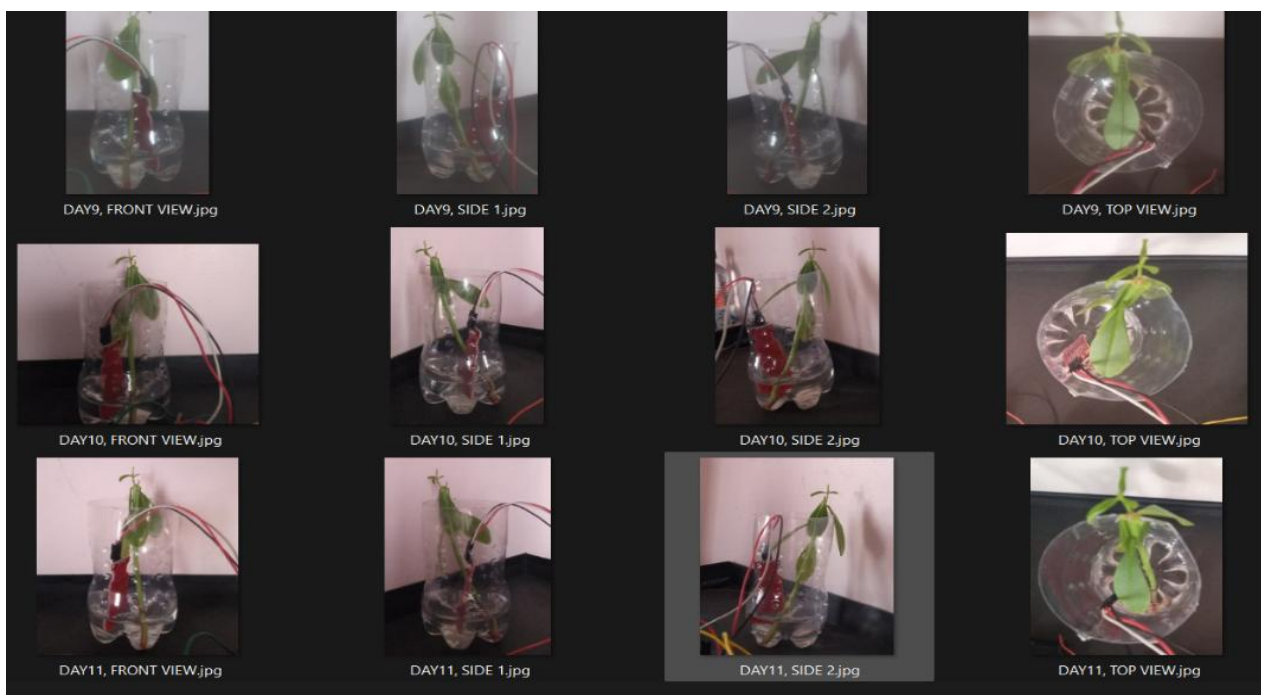


Figure 8: Dataset Samples from the TT-GAN Dataset.

3) Dataset Storage and Publication: Each image was saved in JPG (.jpg) format and organized within a folder structure. The complete dataset was uploaded to a public repository on Kaggle accessible for download at charlesekanem.com.ng/datasets.html.

E. VAE-GAN Architecture Design

The model adopted a hybrid architecture. This design combined the structured latent space learning of VAEs with the adversarial training mechanism of GANs. The architecture consisted of 3 primary components:

- a) Encoder network
- b) Decoder/generator network
- c) Discriminator network.

The encoder operated on RGB images resized to 64×64 pixels and normalized to a fixed intensity range of [1, 1]. Each input image was processed through a convolutional feature extraction pathway that reduced spatial resolution while increasing channel depth. This hierarchical encoding captured visual characteristics of *Talinum triangulare*, including leaf contours, venation patterns and texture variations. Instead of generating a deterministic representation, the extracted features were mapped into a latent probability distribution parameterized by a mean vector (μ) and a logarithmic variance vector ($\log \sigma$). During training, latent vectors were sampled using reparameterization which enabled stochastic sampling while preserving differentiability for backpropagation. This latent representation encouraged continuity within the latent space and improved the model's generalization capability.

The sampled latent vector was projected back into image space through a fully connected expansion layer followed by a sequence of transposed convolutional layers that restored spatial resolution until full-sized 64x64 RGB images were reconstructed. Rectified Linear Unit (ReLU) activation functions were applied between layers for non-linearity and a Tanh activation function was applied at the output layer to bound pixel values within the normalized intensity range of [1,1]. In addition to reconstructing input images, the decoder simultaneously functioned as the GAN generator by producing synthetic images from randomly sampled latent vectors.

The discriminator received both real images from the training dataset and reconstructed images from the decoder. It consisted of successive convolutional layers with progressively increasing feature depth enabling the learning of hierarchical discriminative representations. A binary classification was employed to distinguish real images from generated samples. During training, the variational-generative pathway was optimized alternately. While the discriminator minimized the classification error between real and generated images, the encoder-generator pair minimized a composite objective comprising reconstruction loss, latent-space regularization through Kullback-Leibler divergence and adversarial loss. This joint optimization encouraged image reconstruction, smooth latent representations and the generation of better realistic synthetic images.

Summarized in Table I is the training configuration used for the VAE-GAN architecture. Presented in Figure 9 is an illustration of the VAE-GAN Architecture.

TABLE I
TRAINING CONFIGURATION OF THE PROPOSED VAE–GAN
ARCHITECTURE

Parameter	Value
Image Resolution	64 × 64
Latent Dimension	128
Batch Size	16
Optimizer	Adam
Learning Rate	0.0002
Training Epochs	1000
Generator Activation	ReLU/Tanh
Discriminator Activation	LeakyReLU

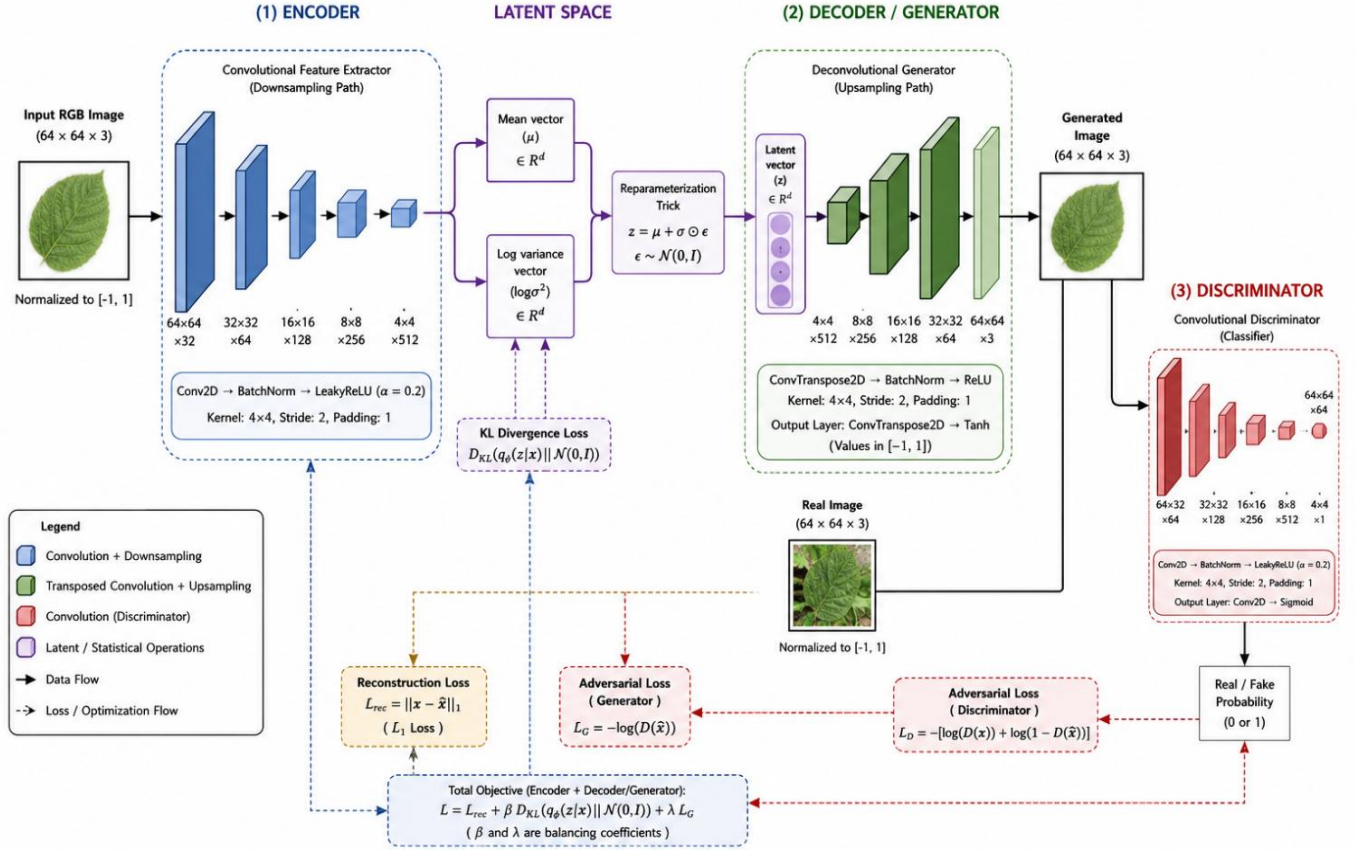


Figure 9: Architecture of VAE-GAN Model consisting of 3 components: (1) Encoder which maps input images into a probability distribution, (2) Decoder/generator which reconstructs images from latent vectors using reparameterization, (3) Discriminator which distinguished real images from generated images using binary classification through adversarial learning.

F. Training Strategy and Loss Functions

The variational autoencoder (VAE) component was trained using a composite loss consisting of a reconstruction loss and a Kullback–Leibler (KL) divergence term. The reconstruction loss enforced pixel-wise similarity between the input image x and its reconstruction $\hat{x}$:

$$L_{\text{rec}} = \sum_{i,j,k} (x_{i,j,k} - \hat{x}_{i,j,k})^2 \quad (1)$$

The KL divergence regularized the latent space by constraining the learned distribution towards a unit Gaussian prior. If the latent vector has dimension d with mean μ_i and standard deviation σ_i , the KL divergence is:

$$L_{\text{KL}} = -\frac{1}{2} \sum_{i=1}^d (1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2) \quad (2)$$

The adversarial (GAN) component employed a discriminator D to distinguish real images from generated samples $G(z)$, where z is a latent vector. The adversarial loss is thus:

$$L_{\text{adv}} = \frac{1}{N} \sum_{n=1}^N \left[\log D(x^{(n)}) + \log(1 - D(G(z^{(n)}))) \right] \quad (3)$$

The total training loss combined all three components:

$$L_{\text{total}} = L_{\text{rec}} + \beta L_{\text{KL}} + \gamma L_{\text{adv}} \quad (4)$$

β and γ are weighting coefficients that balance the latent-space regularization and adversarial realism. Training was performed for 1000 epochs, with intermediate outputs sampled every 10 epochs to monitor convergence and visual quality.

G. Evaluation Metrics

The performance of the VAE-GAN model was measured using both numbers and visual checks. For the numerical evaluation, we looked at how accurately the model could reconstruct images and how the losses changed during training. For the visual evaluation, we examined the generated images and compared them side by side with real plant images. Using both types of evaluation allowed us analyse the model's performance both mathematically and visually.

4. RESULTS AND ANALYSIS

This section presents the experimental results obtained using the TT-GAN framework and hydroponic *Talinum triangulare* dataset. Both qualitative and quantitative analyses are conducted to evaluate the behaviour of the VAE-GAN hybrid model under limited data conditions. The analysis emphasized training stability and the influence of dataset scale on generative performance. The results provided empirical insights into the strengths and limitations of hybrid generative models when applied to small sized botanical image datasets.

A. Training Behaviour and Convergence

Across training epochs, the generator and discriminator losses exhibited stable convergence without evidence of extreme training instability. The reconstruction loss associated with the VAE component decreased steadily which indicated improved learning of latent representations. However, adversarial loss fluctuated throughout training suggesting that while the discriminator had learned to distinguish real from generated samples, its ability to implement fine-grained realism was bounded by the limited dataset size.

B. Qualitative Evaluation of Generated Images

Qualitative evaluation of generated samples by the VAE-GAN model showed that the model progressed from generating total noise to generating faintly noticeable images over the course of 1000 training epochs. Figures 10 through to 20 present generated samples by the model at epochs 1, 100, 200, 300, 400, 500, 600, 700, 800, 900 and 1000 indicating the initial progression of its performance at epoch 100, the improvement at epoch 600 and the final sample at epoch 1000. Noticeable after Epoch 600 was the reintroduction of noise in the samples by the model. This caused the samples to appear more distorted and blurred. The fine-grain details present in the dataset were not captured as well which is behaviour consistent with known limitations of VAE-based reconstruction which often average pixel values in uncertain regions when trained on small datasets. These results demonstrate that the hybrid architecture is more effective in learning coarse structural features even under limited data constraints.

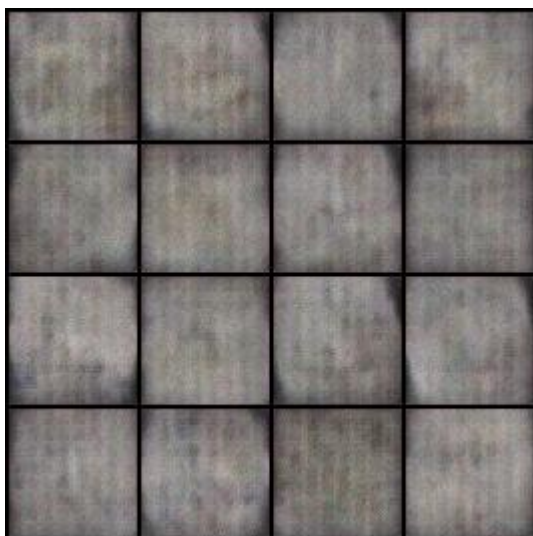


Figure 10: Generated Sample after Epoch 1



Figure 11: Generated Sample after Epoch 100

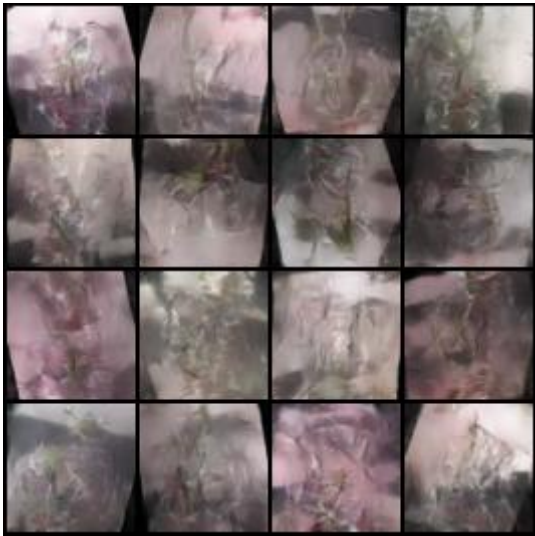


Figure 12: Generated Sample after Epoch 200



Figure 13: Generated Sample after Epoch 300

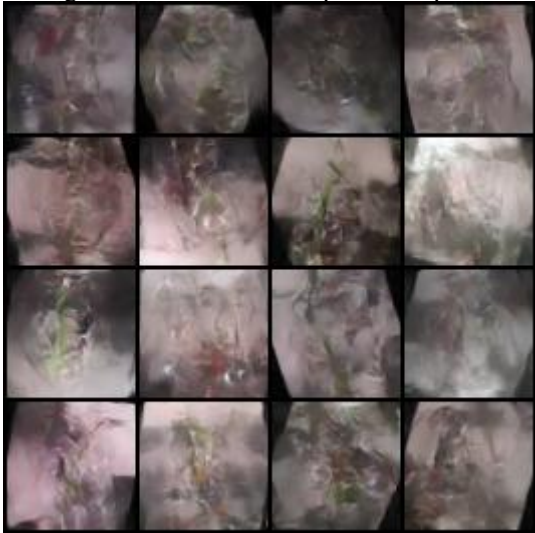


Figure 14: Generated Sample after Epoch 400



Figure 15: Generated Sample after Epoch 500



Figure 16: Generated Sample after Epoch 600



Figure 17: Generated Sample after Epoch 700



Figure 18: Generated Sample after Epoch 800

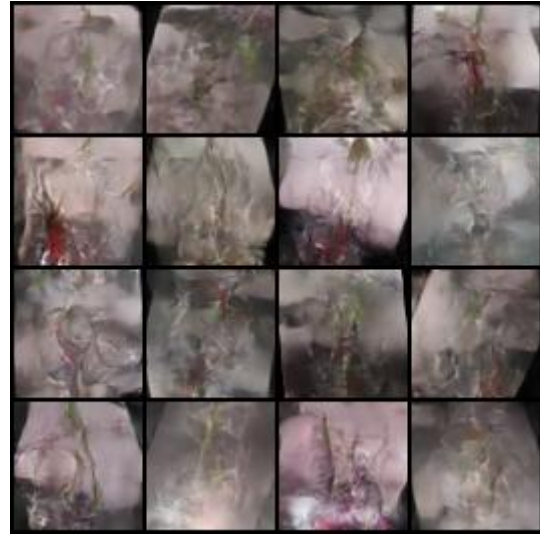


Figure 19: Generated Sample after Epoch 900

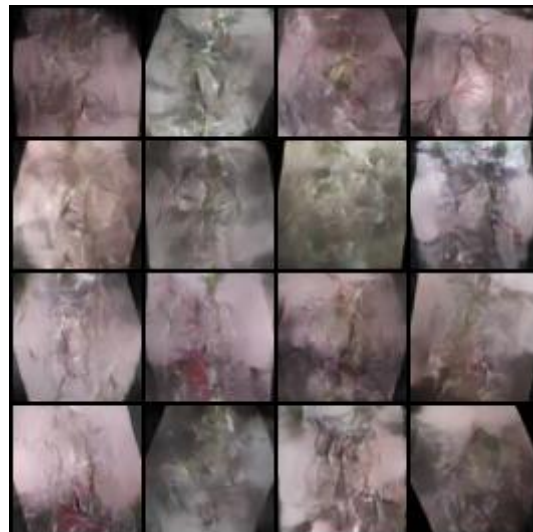


Figure 20: Generated Sample after Epoch 1000

C. Quantitative Image Quality Assessment

<<IMAGES AND MORE WILL BE INCLUDED HERE>>

D. Effect of Dataset Size on Generative Fidelity

The observed results underscore the strong dependence of generative image quality on dataset size in plant-image generation tasks. While the dataset enables the learning of meaningful global plant structures, its limited scale restricts the model's capacity to generalize fine-grained textures. As a result, the VAE-GAN hybrid converges toward visually plausible but perceptually smoothed outputs. These findings highlight an important empirical insight that with small-data, VAE-GAN architectures prioritize structural coherence over texture realism particularly in biological domains such as hydroponic plant imagery.

5. CONCLUSION

This paper introduced TT-GAN, a publicly documented hydroponic image dataset of *Talinum triangulare* and an accompanying evaluation of a hybrid VAE-GAN architecture under limited-data conditions. An IoT-assisted hydroponic cultivation system was developed to support controlled image acquisition and environmental monitoring, resulting in a curated dataset containing 120 images captured over a 30-day growth period.

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